From Mass Spectrum to Molecule

Technical brief · October 18, 2026

How a top public model for Kaggle's Enveda CASMI 2026 challenge works, and what our strategy has found so far

In short. A top public pipeline (reported 0.450) *retrieves* known structures of the right mass, scores them with several kinds of spectral evidence, fuses the rankings and breaks isomer ties with a spectrum-predicting model. Our tracks (Section 3): we forked it unchanged with its pinned software (A, score pending), learned that our novel-molecule generator adds at most ~0.01 (B), found two bugs where test and training data are processed differently (C) and are building a fully rule-compliant pipeline (D).

1 The problem

In **LC-MS/MS**, liquid chromatography separates the sample; a tandem mass spectrometer then ionises each molecule, weighs the intact ion (the **precursor m/z**, mass-to-charge ratio), breaks it by gas collisions (20, 40 or 60 eV) and weighs the charged pieces, the **fragments**. A **spectrum** lists the (fragment m/z, intensity) peaks. **Task:** for each hidden molecule (natural-product-like, i.e. resembling molecules made by organisms; 1–16 spectra, median 3) return up to 25 ranked structures as **SMILES** (structures written as text, e.g. CCO = ethanol).

Why both? The precursor mass, precise to a few ppm, nearly fixes the molecular formula, but many structures share one formula (**isomers**, e.g. ethanol and dimethyl ether, both C_2H_6O). Only the fragment pattern, which depends on how the atoms are connected, tells them apart: shortlist by mass, rank by fragments.

Score, MRR@25 (mean reciprocal rank): mean over molecules of 1/rank of the first correct guess (rank 2 gives 0.5; outside the top 25, 0). Correct = same atoms bonded the same way (first block of the tautomer-canonical InChIKey: 3-D arrangement ignored, tautomers count as equal).

Class	Test molecule…	Share*	Approach
1	has public spectra	~16–28%	library search (MRR ~0.93)
2	known†, no public spectra	~20–45%	retrieve, rank isomers
3	novel: not in PubChem	~39–52%	generate it

*Community estimates; true shares hidden. †In PubChem or COCONUT. Class 3 is hardest: the exact connectivity must be built from scratch.

Data and rules. 2.54M training spectra of ~275,810 structures from 11 public libraries. Code runs offline in a Kaggle notebook (≤ 9 h); prize entries need public, free, commercially usable data (no NIST) and reproducible weights.

A noisy scoreboard. Of ~400 hidden molecules the public board uses ~124 (31%): one molecule moving from miss to rank 1 is worth ~0.008 (standard error ~0.016), so gaps of 1–2 molecules are noise. Final standings use the other 69% (private board). Oct 17: top 0.495, best public notebooks ~0.44–0.45, 3,486 teams.

KEY TERMS

Da (dalton): ≈ mass of one H atom; for these singly charged ions m/z = mass in Da. **ppm**: parts per million (10 ppm of 400 Da = 0.004 Da).

Adduct: the charged form measured, e.g. $[M+H]^+$ or $[M+Na]^+$ (10 allowed); it converts precursor m/z to neutral mass M.

PubChem: public database of ~105M known structures; **COCONUT**: open natural-product database.

Fingerprint: one bit per substructure (present/absent). *Tanimoto* = bits ON in both ÷ bits ON in either (1 = identical, 0 = nothing shared).

Analog: a training molecule with a similar spectrum. **LightGBM**: boosted decision trees; **LambdaRank** trains them to put the right answer near the top of a list.

2 How the current public model works

One of the strongest public notebooks (dmitriigluzdov, "From Spectra to Structures", reports 0.450) stacks community work into a *retrieve-then-rank* pipeline (Fig. 1):

1. **Mass → candidates.** M = median neutral mass over the spectra; candidates = all structures of a ~711k pool (training + COCONUT) within 10 ppm of M (30 ppm fallback): tens to hundreds, mostly isomers (Fig. 2).

2a. **Engine 1** (ahmedberatozer v4n), ~100 features per candidate:
- *library*: peak-matching similarity (spectral entropy) to the candidate's own reference spectra; handles most of class 1;
- *analog propagation*: training molecules with similar spectra (peaks may be offset by their mass difference) vote for candidates that resemble them (Tanimoto);
- *FPNet*: a neural network predicts from the peaks which substructures are present (the fingerprint); agreeing candidates score high;
- *fragmentation*: peak intensity explained by breaking 1–2 bonds, plus a learned fragment scorer;
- *generator*: chemical edits of the 6 closest spectral analogs (class 3).

LightGBM rankers combine the features; final score = z + 0.15 × popularity (z: score standardised per candidate list; popularity: log of PubChem substance + literature counts, since well-studied compounds are likelier answers).

2b. **PubChem channel**, for class-2 molecules outside the pool: fingerprint models score PubChem structures (of ~105M) in the mass window (z + 0.25 × popularity). Top picks are interleaved as a hedge at ranks 2, 4, …, 10 (or 4, 8, …, 20), promoted to rank 1 if the fingerprint-score margin is large or the pick well documented, skipped given a library match ≥ 0.9.

2c. **Engine 2**: an independent second opinion (own candidates, channels, two boosted rankers); combining two opinions reduces errors.

3. **Fusion:** reciprocal-rank fusion (each list adds 1/(c + rank), c a constant); duplicates merged.

4. **Re-rank isomers:** in each formula group, forward models ICEBERG/GLACIER predict each candidate's spectrum; the best match to the measured one moves up (fragment score as fallback). Engine 2's top pick goes to rank 2.

History: ~0.34 (mid Sep) → 0.40 (engine fusion) → 0.44 (forward models) → 0.45 (tuned PubChem promotion, deeper engine-2 trees); the last +0.01 ≈ one molecule, within noise.

Weak spots: almost no class-3 hits; many misses are the *wrong isomer*; tweaks accepted on the noisy board without cross-validation; some weights have non-commercial licences (barred for prizes) or unpublished construction.

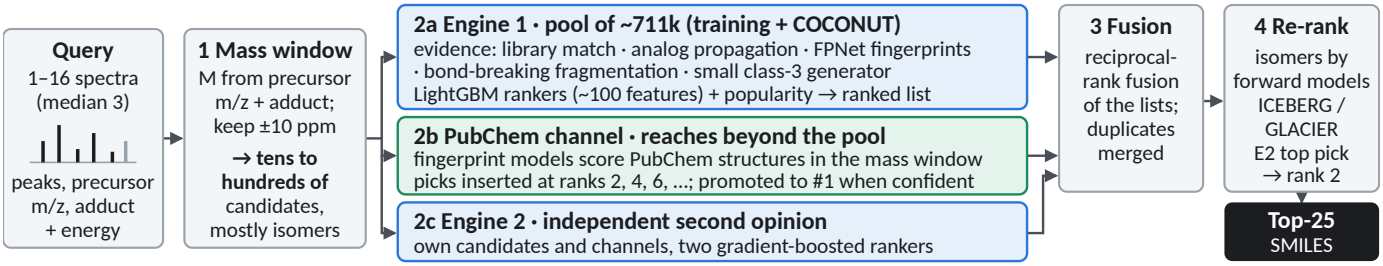


Figure 1. The public model as a pipeline (numbers match the steps in Section 2). The blue and green boxes generate and rank candidates independently and decide *which* structures (mostly known ones) enter the list; step 4 only reorders candidates that share a formula (isomers), where mass cannot help.

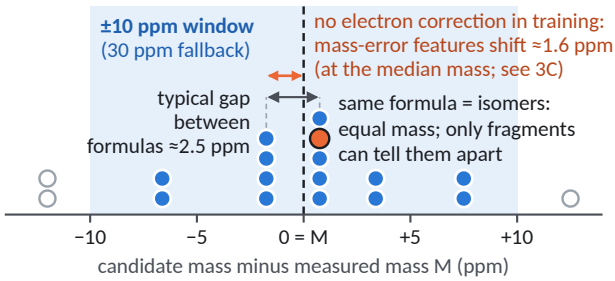


Figure 2. Mass window (schematic; Sec. 2, step 1). Dots = structures, columns = formulas, hollow = outside (never scored). The true one (orange) sits near, not at, M (measurement error); ranking *inside* its isomer stack decides the score. Orange arrow: a feature shift, not a window shift (3C).

3 Our strategy (first session, Oct 17–18)

A Get on the board. Forked the 0.450 notebook unchanged on its fixed software versions (copies on newer ones lost ~0.04). Score pending.

B Class-3 deep-dive (the largest untapped share).

- **Proxy:** 16,917 training structures are not in PubChem (all 106M keys checked): real measured molecules, a class-3 stand-in. 45% split at one amide/ester bond into two PubChem-known pieces (N-acyl amino acids, bile-acid conjugates, fatty-acid esters).
- **NovelEngine:** two generators; both output only structures absent from PubChem, COCONUT and training.
  - *Analog derivatives*: find training spectra similar to the query by modified cosine (which also matches peaks shifted by the mass difference Δm), then edit the analogs' *structures* with transformations (hydroxylation, methylation, glycosylation, chain extension, …) whose mass change equals Δm.
  - *Conjugates* (motivated by the 45%): known acid + known amine/alcohol joined by an amide/ester bond (or glycoside), whose masses minus the water lost on bond formation equal M; ranked by fragment ions typical of each half.
- **Site localisation** (our new idea, Fig. 3) scores where on the analog the edit sits; a **learned ranker** (LambdaRank) re-orders the candidates (results: Section 4).
- **Reality check:** only-novel candidates scored 0.010 on the public board (about one molecule at rank 1) vs ~0.045 predicted from the proxy (Fig. 5): reach was overstated ~4–5×, so our class-3 engine adds at most ~0.01. Generating is far harder than retrieving: even on the proxy, analog derivatives reached the right structure for only 24% of molecules.

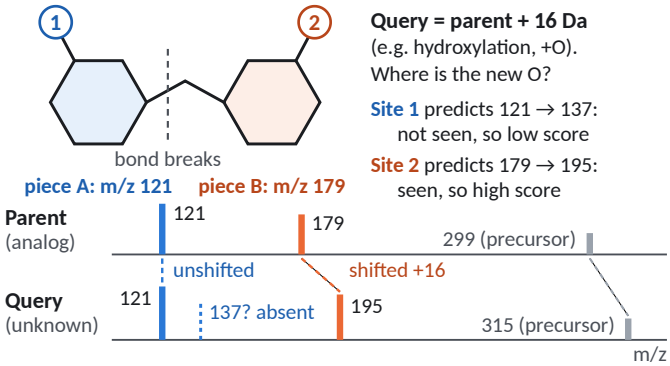


Figure 3. Site localisation (toy numbers). Peaks reappearing unshifted come from pieces without the modification, peaks shifted by Δm from pieces carrying it; sites are scored by this consistency (here: site 2).

C Bug hunt on the public pipeline. Two *train/serve mismatches* (a feature computed one way in training, another way at test time, so learned rules misfire): (1) number of peaks: raw test spectra (median 230) vs cleaned training ones (median 52); (2) positive ions lack one electron (0.000548 Da): hidden-test m/z values are corrected for it, but the enveda-180 training spectra used to fit all models are not, so mass-error features are off by ~1.6 ppm at the median mass, two-thirds of the ~2.5 ppm typical gap between formulas (Fig. 2). Two patched variants: pending.

D Compliance-first pipeline of our own (all code and weights ours, reproducible, permitted data).

- **Pool** of 726,198 structures (training + COCONUT Oct-2026). **Channels:** library entropy similarity, analog propagation, fragment explanation, COCONUT documentation priors (organisms, literature, synonyms) and our own model.
- **Dual encoder** (Fig. 4): learns to place the true structure closer than up to 64 same-mass "hard negatives" (contrastive loss) and to predict fingerprint bits. Its held-out score (Sec. 4) is from an easier task than the leaderboard (the answer is always in the list): not comparable with 0.450. Longer runs in progress.
- **Ranker:** LambdaRank on simulated queries from 8,000 training structures (class 1: some of a structure's spectra are the query, the rest stay in the library; class 2: all removed), validated on 1,500 disjoint structures (250 Enveda natural products measured like the test, 750 other natural products, 500 others), using the *measured* mass (the true one would leak the answer) and electron correction.

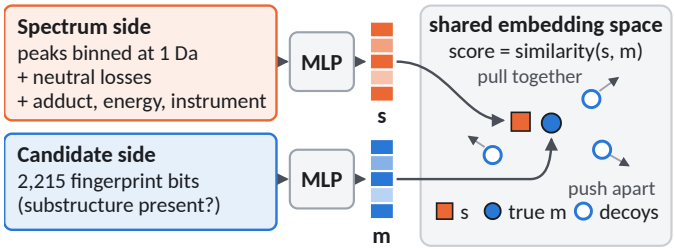


Figure 4. Our dual encoder. Two MLPs (small fully connected networks) map spectrum and candidate to vectors s and m; training pulls true pairs together and pushes decoys apart; candidates are ranked by similarity.

4 Results so far & lessons

	Experiment (measured on)	Result
A	Fork of the 0.450 notebook (board)	*pending*
B	Analog-derivative reach (proxy)	24%
B	Site localisation (derivative MRR, proxy)	0.061 → 0.107
B	Learned ranker (proxy MRR)	0.124 → 0.269
B	Novel-only / known-only probes (board)	0.010 / 0.085
C	Two bug-patched variants (board)	*pending*
D	Dual encoder, 1 CPU epoch (same-mass ranking task; not a leaderboard estimate)	MRR 0.51, top-1 36%

Proxy, held-out and board numbers come from different tasks; only board rows compare with 0.450; the two proxy MRRs are not cumulative.

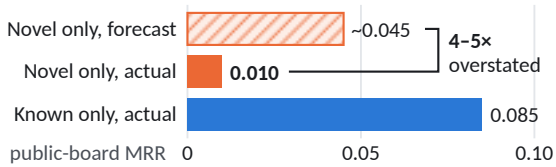


Figure 5. Class-3 probe, public board. Novel-only candidates scored 0.010 (about one molecule at rank 1) vs ~0.045 forecast; known-only candidates from the same engine scored 0.085.

TAKEAWAYS

- **A proxy is a hypothesis about the test: validate it against the leaderboard early.** The board is noisy (±0.016), so offline validation that mimics the test is also needed.
- With current methods the reachable points are in classes 1–2: rank the right *isomer* first.

5 Next steps (track E)

Compare variants on the board; finish our own pipeline (target ≥ 0.45, fully compliant); add the class-3 engine only where the main list is weak; choose the two submissions scored on the private board by validation + board. **Why:** hosts "reserve the right to check all submissions"; if enforced, strong compliant entries may outrank non-reproducible ones.